



PRODUCT REQUIREMENTS DOCUMENT v2.0

Bali YTTC

Complete Digital Ecosystem

Client	Bali Yoga Teacher Training Center
Website	baliyttc.com
Location	Ubud, Bali, Indonesia
Prepared by	Adruva Solution
Version	v2.0 — Final Discovery
Date	May 2026
Testing	Vercel + Railway (VPS post client confirmation)
Status	Ready for Development

5 PRODUCTS IN THIS ECOSYSTEM

- 01 Marketing Website — baliyttc.com (17 pages, 8 languages, SEO-first)
- 02 Student Web App (PWA) — Enrolled students only, post-payment access
- 03 Admin Panel — Owner + 6 staff roles (RBAC), full control center
- 04 Blog CMS — AI-assisted, multilingual, deep-dive SEO content
- 05 Support Bot — Rule-based FAQ, website visitors, 8 languages

01 Project Overview

Aduva Solution is developing a complete custom digital ecosystem for Bali Yoga Teacher Training Center — a Yoga Alliance-certified school in Ubud, Bali. The project replaces the existing WordPress site with a fully custom, high-performance, multilingual platform built from scratch. Testing will be done on Vercel + Railway before any production deployment.

Client	Bali Yoga Teacher Training Center (Bali YTTC)
Website	baliyttc.com
Location	Ubud, Bali, Indonesia
Certification	Yoga Alliance registered
Courses	100hr, 200hr, 300hr, 50hr Hatha-Vinyasa, Retreats, Workshops
Target Audience	International students — primarily European + English-speaking
Current Platform	WordPress + Elementor (being replaced fully)
Testing Environment	Vercel (frontend) + Railway (backend + DB)
Production	VPS to be decided post client confirmation

Ecosystem — 5 Products

#	Product	Audience	Core Purpose
01	Marketing Website	Public visitors	SEO-first, courses, enrollment, blog, bot
02	Student Web App (PWA)	Paid + enrolled students only	Lessons, schedule, progress, certificates, alumni
03	Admin Panel	Owner + 6 staff roles	Full control center for all 5 products
04	Blog CMS	Admin / SEO Editor	AI-assisted multilingual long-form SEO content
05	Support Bot	Website visitors only	Rule-based FAQ, 8 languages, 24/7

02 Technology Stack

Frontend

Layer	Tool	Purpose
Framework	Next.js 14 (App Router)	React-based, SSR + SSG, SEO-friendly, fast builds
Styling	Tailwind CSS	Utility-first, mobile-first, consistent design system
UI Components	Shadcn/ui + Radix UI	Accessible, unstyled base components
Multilingual	next-intl	Subfolder routing /de/ /fr/ /es/ etc, auto-detect
Fonts	Google Fonts (max 2 families)	Free, fast, self-hostable — client to choose
Icons	Lucide React	Lightweight SVG icons
Animations	Framer Motion (minimal)	Subtle transitions only — performance priority
Forms	React Hook Form + Zod	Client-side validation, lightweight
PWA	Next.js PWA plugin	Installable on phone, no App Store needed
Video	YouTube Unlisted + Signed Embed	Free, CDN-fast, domain-locked playback

Backend

Layer	Tool	Purpose
Runtime	Node.js 20 LTS	Stable, fast I/O, widely supported
API Framework	Express.js	Lightweight REST API
Validation	Zod	Schema validation on every API endpoint
ORM	Prisma	Type-safe DB queries, migration management, no raw SQL
Authentication	Firebase Auth (free tier)	Google one-tap + email/password, unlimited MAU
Session / JWT	Custom JWT — httpOnly cookies	Secure, XSS-protected token storage
File Uploads	Multer	Profile photos, gallery images, course materials
PDF Generation	PDFKit	Auto-generate certificates on admin approval
Cron Jobs	node-cron	Reminders, batch alerts, abandoned enrollment checks
Email	Brevo + Resend	Brevo = newsletters, Resend = transactional (OTP, confirm)
Payments	Stripe + PayPal SDK	International cards + PayPal, EUR primary + USD

Database, Cache & Infrastructure

Layer	Tool	Purpose
Database	PostgreSQL 16	Relational, ACID, Row Level Security (RLS) built-in
ORM / Migrations	Prisma	Type-safe, version-controlled schema migrations
Row Level Security	PostgreSQL RLS policies	DB-level access — student sees only own rows

Layer	Tool	Purpose
Cache	Redis 7	API response cache, sessions, rate limit counters
Search	PostgreSQL Full-Text Search	Blog + course search, no extra service needed
Testing — Frontend	Vercel	Auto-deploy on git push, free tier, Next.js native
Testing — Backend	Railway	Node.js + PostgreSQL + Redis, free tier available
CI/CD	GitHub Actions	Auto-deploy on push to main branch
Monitoring	UptimeRobot (free)	Uptime alerts if site goes down
Error Tracking	Sentry (free tier)	Real-time error alerts frontend + backend
CDN	Cloudflare Free	Static assets, DDoS protection, SSL, WAF

Third-Party Services

Service	Tool	Cost	Purpose
Auth	Firebase Auth	Free (unlimited)	Google + email login
Email Transactional	Resend.com	Free (100/day)	Enrollment confirm, OTP, cert, reminders
Email Marketing	Brevo	Free (300/day)	Newsletter, early bird announcements
Payments	Stripe	2.9% + 30c/txn	Card payments — EUR + USD
Payments	PayPal	~3.49%/txn	PayPal wallet — popular in Europe
Video	YouTube Unlisted	Free	Course videos — signed embed only
Analytics	Google Analytics 4	Free	Traffic, behavior, conversions
Push Notifications	Web Push API (PWA)	Free	Browser push — no third party
Reviews	Google + TripAdvisor	Free	Embedded reviews + auto review request
Bot	Rule-based (custom)	Free	FAQ bot — no AI API cost
Maps	Google Maps Embed	Free tier	School location on contact + about page

03 Marketing Website — baliyttc.com

Custom Next.js build from scratch. Mobile-first, SEO-first, 8-language subfolder structure. All 17 pages below.

Pages (17 Total)

Page	Key Content & Features
Homepage	Full-screen cinematic video hero (client footage), USP, course cards, interactive stats (admin-managed), video testimonials, gallery preview, blog preview, countdown batch widget, CTA strip
100hr YTT	Interactive day-by-day syllabus, batch dates + live seat counter + countdown timer, accommodation selector, deposit or full payment, teacher profile, FAQs, early bird badge
200hr YTT	Same structure as 100hr
300hr YTT	Same structure — advanced specifics
50hr Hatha-Vinyasa	Short course page — same structure
Retreats	3-day + 7-day pages (Ubud + Canggu)
Workshops	Sound Healing, Acro Yoga, Arm Balancing, Mandala
About / School	Story, mission, Yoga Alliance cert, Ubud location, custom Google Maps embed with landmarks
Instructors	Teacher profile cards — admin managed, high-quality photos
Pricing	Admin-managed table — seasonal pricing, early bird badges, deposit info
Gallery	Professional photos (owner upload) + student-submitted photos (admin approved)
Testimonials + Alumni	Text + photo + video testimonials, Google Reviews embed, alumni stories merged
Blog	Long-form 2000+ word articles, category filter, search, multilingual
FAQ	Accordion — admin managed, multilingual
Visa Information	Dedicated page — visa process for European students
Contact	Form + WhatsApp button (opens owner WhatsApp) + Google Maps + email
Terms & Policy	Static — GDPR compliant, cookie consent

Website — Key Features

- Multilingual: 8 languages — EN, ES, DE, KO, ZH, JA, FR, RU via subfolder routing (/de/, /fr/ etc)
- Auto language detection: IP + browser detect, manual switcher in navbar
- SEO: Schema markup (Organization, Course, Article), sitemap auto-update, Open Graph, Twitter Cards
- Rich search results: Batch availability schema — Google shows 'Next batch: March 15 — 3 seats left'
- Interactive syllabus: Day-by-day clickable per course page
- Live batch widget: Next batch date + seat count + countdown timer on every course page
- Seasonal pricing: Admin sets different prices per season per course
- Early bird pricing: Admin sets price + deadline, auto-switches after
- Discount coupons: % or fixed, usage limit, expiry date
- Enrollment flow: Course → batch → accommodation → coupon → deposit or full → PayPal/Stripe → app access
- Accommodation: Admin configures room types + prices per course (currently Single + Shared)

- Currency: EUR primary + USD + auto-detect by country, manual toggle
- Abandoned enrollment recovery: Admin toggles on/off — automatic email reminders
- Lead magnet: Free PDF guide download → email capture → Brevo newsletter sequence
- Early bird email: Auto-announce new batches to newsletter subscribers with PDF guide
- Failed payment: Email to admin + student both with retry link
- Visa info page: Dedicated page + automatic email to student post-enrollment
- WhatsApp button: Opens owner WhatsApp directly
- Student photo gallery: Submit via app → admin approve → website gallery
- Review request: Auto post-course email — Google + TripAdvisor links
- GDPR: Cookie consent banner, privacy policy, data deletion on request
- Support bot: Bottom-right bubble, rule-based FAQ, 8 languages
- Social proof numbers: Manually updated by admin (graduates, years, rating)
- Comparison/guide pages: Client decides content strategy (high-converting SEO pages)

04 Student Web App (PWA)

Progressive Web App — opens in browser, installable on phone home screen. No App Store needed. Access strictly tied to payment-confirmed email. Firebase Auth login must match enrollment email exactly.

Access Rules

Who can access	Only students with confirmed payment. Visitors cannot reach any PWA route.
Email lock	Login email must match the email used during payment — system verifies this automatically
Login methods	Google one-tap OR email + password (Firebase Auth)
Pre-arrival access	After deposit payment — limited access: prep videos, course manual, Bali travel guide, packing checklist
Full access	After full payment — complete PWA access
Multiple courses	Same account supports multiple enrollments across different courses
Alumni access	Post-course: limited — certificate download + school updates feed
Alumni discount	Admin toggles on/off — sets discount amount — auto-applies to alumni at checkout
Token storage	httpOnly cookie — protected from XSS, not localStorage
Session	8 hours — auto logout after inactivity

Features — Full Breakdown

Feature	Detail
Dashboard	Welcome message, today's class, progress summary, quick links to all sections
Pre-Arrival Portal	Unlocked after deposit — prep yoga videos (YouTube embed), course manual PDF, Bali travel guide, packing checklist
Video Lessons	YouTube Unlisted embedded, organised by module/week. Signed URLs — domain-locked, cannot be shared externally
Class Schedule	Weekly + daily calendar. Teacher name, time, yoga style, room. Balinese ceremony dates auto-blocked (admin marks them)
Progress Tracker	Hours completed vs required, module completion %, visual progress bar
Announcement Board	School-wide — all enrolled students. Teacher posts updates, students react (emoji) + simple reply. No threaded chat.
Personal Notes	Private notepad — auto-save, only student sees, not visible to teachers or admin
Profile	Name, photo upload, course, batch, enrollment date, payment status, certificate status
Profile Photo	Student uploads photo — shows on profile and announcement board next to name
Certificate Download	PDF auto-generated after admin manually approves. Yoga Alliance format — design by client.
Review Section	Student can leave Google + TripAdvisor review directly from portal with one click

Feature	Detail
Testimonial Submit	Student writes review → admin approves → published on website testimonials
Notifications	Simple on/off toggle — Email + browser push (PWA built-in, no third party)
Alumni Feed	Post-course: school updates + announcements visible. Certificate download available permanently.
Visa Info	Dedicated info section with Bali visa process for European students

05 Admin Panel

Secure owner-only dashboard with delegated staff roles. Single control center for all 5 products. Login requires email + password + TOTP 2FA (Google Authenticator).

Admin Modules — Complete List

Module	What Can Be Done
Student Management	View all students, approve/reject enrollment, revoke/grant app access, view individual progress, send direct email, manage alumni status
Course & Batch Management	Create/edit/delete courses, set regular + seasonal + early bird pricing per course, create annual batches, one-click clone to next year (prices + teachers carry over), set accommodation options per batch, close batch manually, view waitlist
Accommodation Settings	Add/edit/remove room types (Single, Shared — extendable), set price per room per course, mark as mandatory or optional
Ceremony Calendar	Mark Balinese ceremony dates — automatically blocks 'No class' in student schedule. Admin adds dates annually.
Pre-Arrival Content	Upload prep videos (YouTube links), course manual PDF, Bali travel guide, packing checklist — per course
Payments	View all Stripe + PayPal transactions, payment status per enrollment (deposit/full), failed payment alerts, issue refunds (manual approval), download invoices
Coupons & Discounts	Create coupons (% or fixed amount), set expiry + usage limit, early bird automation, alumni discount toggle + amount, view usage stats
Abandoned Enrollment	Toggle on/off — when on: auto email at 1hr, 24hr, 3 days after incomplete payment
Email Templates	Simple text editor — owner edits text content. Layout fixed by developer. Templates: enrollment confirm, pre-arrival, reminder, certificate, review request, early bird announcement, visa info
Blog CMS	Rich editor, AI improve + SEO suggest + market insights, meta fields, schedule publish, categories, multilingual (8 languages)
Gallery	Upload professional photos, approve/reject student-submitted photos, delete, reorder
Testimonials	Approve student submissions, add manually (text + photo), video testimonial embed links, Google Reviews toggle
Teacher Profiles	Add/edit/delete teacher profiles, photo, bio, specializations — admin only (teachers cannot self-edit)
Instructor Accounts	Invite teacher via email → limited login access for schedule + announcements only
Support Bot	Update FAQ knowledge base, view recent conversations, see unanswered questions, toggle bot on/off
Analytics Dashboard	Total enrollments (monthly/yearly), revenue (monthly/yearly), per-course enrollment breakdown, coupon usage stats, referral stats (if enabled later), pending payments count, Google Analytics embed
Social Proof Numbers	Manually update: total graduates, years of experience, average rating — shown on homepage
Notification Management	Send manual push/email to all students or specific batch



Module	What Can Be Done
Staff Management (RBAC)	Invite staff via email, assign role, deactivate account, change role, view audit log

RBAC — 6 Staff Roles

Role	Access Summary	Cannot Access
Super Admin (Owner)	Everything + staff management. Only role with full system access.	Nothing blocked
Student Manager	Students, enrollments, app access, certificates, announcements, testimonials, partial analytics	Payments, blog, pricing, staff management
SEO Editor	Blog, SEO fields, bot knowledge base, gallery, testimonials approval, partial analytics	Student data, payments, batches, staff management
Finance Manager	Payments, refunds, coupons, revenue analytics, alumni discount settings	Student personal data, blog, batches, staff management
Course Manager	Courses, batches, seasonal pricing, accommodation, ceremony calendar, clone feature, partial analytics	Payments, student personal data, blog, staff management
Teacher	Own schedule view, post announcements to school board, view own batch students (names only)	All other modules — finance, blog, other batches, staff

All admin actions logged in audit_log table — who did what, when, old value, new value. Super Admin can view full audit history.

06 Blog CMS

Built into admin panel. Admin writes in English — same 8-language translation system covers blog automatically. AI assists but never auto-writes — admin is always the author.

Feature	Detail
Rich Text Editor	Bold, italic, H1-H4, lists, images, embeds, blockquotes, code blocks, tables
AI Writing Assistant	Improve paragraph, better phrasing, SEO keyword suggestions, market-based content insights, readability score — admin writes, AI improves
SEO Fields	Meta title, meta description, slug editor, focus keyword, Open Graph preview, Twitter Card preview
Schema Markup	Auto Article schema — Google rich results
Auto Sitemap	sitemap.xml auto-updates on every publish
Categories + Tags	Organise by yoga style, topic, audience level
Schedule Publishing	Write now, auto-publish at set date + time
Featured Image	Upload + auto-compress and resize for web
Draft + Preview	Save draft, preview live, one-click publish
Multilingual	Same 8-language system as site — translation handled by next-intl
Content Strategy	Long-form 2000+ word deep-dive guides for AI citation + Google authority

07 Support Bot

Feature	Detail
Type	Rule-based FAQ bot — fixed questions + answers. No AI API cost.
Placement	Bottom-right chat bubble — user clicks to open. No auto-popup. Clean, non-intrusive.
Audience	Website visitors only. Enrolled students use the PWA.
Languages	All 8 languages — auto-switches based on detected site language
Knowledge Base	Full website content: all courses, pricing, batch dates, accommodation, FAQs, visa info, location, contact, basic payment info (Stripe + PayPal)
Typical Questions	Which course suits me? What is included? How to book? Refund policy? How to get to Ubud? Visa requirements?
Escalation	Cannot answer → 'Would you like to contact us?' → routes to contact form
Admin Control	Owner updates FAQ knowledge base from admin panel — no developer needed
Conversation Logs	Admin sees recent conversations, identifies unanswered questions, improves knowledge base
Toggle	Admin can turn bot on/off from admin panel

08 Enrollment & Payment Flow

Step	Action	Detail
1	Course page	Student lands on course page — sees interactive syllabus, batch dates, live seat count, countdown timer
2	Select batch	Chooses available batch — seat availability shown live, full batches auto-closed
3	Select accommodation	Room type options shown — Single or Shared (admin-configured). Price breakdown updates live.
4	Apply coupon	Optional — coupon code entered, discount applied instantly
5	Choose payment	Deposit only (admin-set per course) OR full amount. Currency auto-detected (EUR/USD).
6	Pay	Stripe (card) or PayPal — both supported. Failed payment → email to admin + student with retry link.
7	Pre-arrival access	Deposit confirmed → Pre-Arrival Portal unlocked immediately (prep videos, manual, travel guide)
8	Confirmation email	Automated via Resend — booking details, visa info, WhatsApp contact, what to bring
9	Full PWA access	Full payment confirmed → complete Student App access granted
10	Admin notified	Admin panel shows new enrollment, payment status auto-updated, notification bell
11	Pre-course reminders	Auto emails: 30 days before (packing guide), 7 days before (final checklist), 1 day before (see you tomorrow)
12	Certificate	Post-course: admin manually approves → PDF auto-generated → student downloads from PWA
13	Alumni	Course complete → Alumni status → limited PWA access (certificate + updates + review section)
14	Review request	Auto email post-course → Google + TripAdvisor review links

09 Security Architecture

Layer	Tool / Method	What It Does
HTTPS	Cloudflare SSL + HSTS	All traffic encrypted, HTTP auto-redirects to HTTPS
Firebase Token	Firebase Admin SDK	Verifies user identity on every API request — step 1
JWT	Custom JWT — httpOnly cookie	Role-based token issued after Firebase verify — step 2. Not localStorage — XSS safe.
Email Lock	Enrollment email check	PWA access only if login email matches payment email — system enforced
Role Middleware	Express middleware	Every API route checks JWT role before processing — step 3
Database RLS	PostgreSQL RLS policies	Student sees only own rows — DB enforces this even if API has a bug
Input Validation	Zod schemas	Every endpoint validates incoming data type and shape
Rate Limiting	express-rate-limit + Redis	Per IP + per user — prevents brute force and abuse
SQL Injection	Prisma ORM	No raw SQL — all queries parameterised through ORM
CORS	Express CORS config	Only baliyttc.com and subdomains can call API
Admin 2FA	TOTP — Google Authenticator	Admin login requires 6-digit OTP in addition to password
DDoS	Cloudflare WAF	Blocks malicious traffic before it reaches server
Audit Log	PostgreSQL audit_log	Every admin action: who, what, when, old value, new value
Teacher Role	Limited Firebase + JWT	Teacher login scoped to own schedule + own batch announcements only

User Access Summary

User Type	Who	Can Access	Cannot Access
Visitor	Anyone on website	Public pages, blog, bot, inquiry, enrollment + payment	Student PWA, admin panel
Student	Paid + enrolled	PWA — own data only (RLS enforced at DB level)	Other students' data, admin panel
Alumni	Course completed	Certificate download, school updates, review section	Full PWA features, other students' data
Teacher	Invited by owner	Own schedule, own batch announcements	Finance, other batches, student personal data
Staff	Invited by owner	Own role's modules only	All modules outside their role
Super Admin	Owner only	Everything	Nothing blocked

10 Design Direction

Mobile-first, minimal, meditative. Inspired by Bali's natural environment. Premium feel without visual noise. Silent luxury approach — generous whitespace, every element intentional.

Design Principles

- Mobile-first — designed for phone screens first, scales to desktop naturally
- Silent luxury — generous whitespace, nothing on screen without purpose
- Content-rich for SEO — detailed HTML beneath minimal visual layer — user gets peace, Google gets data
- No heavy animations — subtle transitions only, performance always priority
- Accessible — WCAG AA contrast, keyboard navigable, screen reader friendly
- Authentic photography — client to provide Bali shala footage and photography

Recommended Color Palette — Earthy Luxe

Name	Hex Code	Usage
Deep Brown	#2C1810	Primary text, headings, navbar background
Terracotta	#C4714A	CTA buttons, accent elements, highlights
Sage Green	#8B9E6E	Nature element, secondary accents
Gold	#D4A853	Premium labels, testimonial authors, badges
Warm Cream	#FAF7F2	Page background, section backgrounds
Muted Brown	#9A8F87	Body text, captions, secondary content
Pure White	#FFFFFF	Cards, modal backgrounds, forms

Font Pairings — 3 Options for Client to Choose

Option	Heading Font	Body Font	Vibe
Pairing 1	DM Serif Display — italic, weight 300	DM Sans — light 300	Meditative, editorial, warm premium
Pairing 2	Fraunces — italic, optical, weight 300	Figtree — light 300	Soulful, human, approachable warmth
Pairing 3	Nunito Sans — weight 300 only	Nunito Sans — weight 400	Ultra minimal, calm, zero distraction

All fonts from Google Fonts — free, fast, self-hostable. Maximum 2 font families loaded. Client to confirm choice before design phase begins.

11 Development Phases & Timeline

Phase	Duration	Deliverables
Phase 0 — Sign-off	1 week	Contract signed, 50% advance, font + hero confirmed by client, media assets (video/photos) provided
Phase 1 — Design	2-3 weeks	Wireframes all pages, design system, high-fidelity mockups (mobile + desktop), client approval round
Phase 2 — Backend	3-4 weeks	DB schema, Firebase Auth, JWT middleware, RLS policies, Stripe + PayPal, enrollment flow, email system, admin panel core modules
Phase 3 — Frontend	3-4 weeks	All 17 website pages, student PWA, admin panel UI, blog CMS, interactive syllabus, batch widgets
Phase 4 — Integrations	1-2 weeks	Support bot, multilingual (next-intl), Google Analytics, push notifications, review system, pre-arrival portal
Phase 5 — Testing	1-2 weeks	Full QA on Vercel + Railway, content migration from WordPress, cross-device testing, performance audit
Phase 6 — Launch	TBD	Domain + VPS setup (post client confirmation), Cloudflare config, go live on baliyttc.com

Important — Client Action Required Before Phase 0

- Confirm font pairing choice (3 options presented separately)
- Confirm homepage hero style (full-screen cinematic video confirmed — footage needed)
- Provide Bali shala video footage + photography for website
- Provide exact bank/Wise details if bank transfer added later
- Confirm certificate design + content (Yoga Alliance format — client designs template)
- Confirm accommodation room types, pricing per course
- Confirm which European languages to prioritise first in Phase 2 (based on GA data post-launch)

This document represents the complete agreed scope as of May 2026. Any additions post sign-off will be scoped and quoted separately. Prepared by Aduva Solution.